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The Rise and Demise of the Convertible Arbitrage Strategyⁱ

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The Rise and Demise of the Convertible Arbitrage Strategy

Abstract

This paper analyzes convertible arbitrage, one of the most successful hedge fund strategies. The aim of the strategy is to exploit underpricing of convertible bonds by taking a long position in a convertible and a short position in the underlying asset. We find that convertible bonds are underpriced at the issuance dates. At the same time, short sales of underlying equity significantly increase. Both effects are stronger and more persistent for equity-like than for debt-like convertibles. Furthermore, we find that short sales pressures negatively affect stock returns around announcement and issuance dates of convertibles. In our opinion, this contributed to the shift towards issuing more debt-like convertibles in recent years, which in turn substantially lowered the returns from convertible arbitrage.

1. Introduction

Convertible arbitrage¹ has been one of the most successful hedge fund strategies at the turn of the Millenium. Even though the returns of convertible arbitrage have strongly decreased in recent years, its trades currently represent more than half of the secondary market trading in convertible securities with hedge funds as the most important player in this market.² Moreover, in the market of primary convertible bond issues 70% to 75% of the issues are bought by hedge funds (see for example Arshanapalli, Fabozzi, Switzer, and Gosselin, 2004 and Lian, 2006). This implies that hedge funds engaging in convertible arbitrage³ are important investors, both in primary and secondary markets. This paper studies the different effects that convertible arbitrage has on the capital market and its participants.

The Convertible Arbitrage Index HEDG CA that is tracked by CSFB/Tremont shows that annual returns on the convertible arbitrage have for the most part been above 15% in the 1990s and up to 2001. However, the convertible arbitrage performance has deteriorated in later years. The popular press provides different explanations for this, ranging from stable equity markets, rising interest rates, withdrawals from funds, to increased competition in the hedge fund industry and lower volatilities in the main capital markets. However, we demonstrate that the structure of the convertible bond (being either debt- or equity-like) is an important additional explanatory factor to be considered, as it 1) affects the degree of underpricing, 2) affects the “hedging” position in the underlying stock, and 3) affects the excess returns on the issuer’s equity. We show that through this channel convertible arbitrage creates significant economic losses for the issuing companies in particular in the case of equity-like convertibles.

Research on hedge fund inflows and performance (see for example Agarwal, Daniel, and Naik, 2004 and Choi, Getmansky, and Tookes, 2008) has shown that inflows into hedge funds

¹This strategy is aimed at exploiting the underpricing of convertible bonds. It consists of taking a long position in the convertible bond and a short position in the underlying asset into which the bond can be converted. The underlying asset is typically a stock of the company that issues the convertible bond.

²See for example Lhabitant (2002).

³The term convertible bond arbitrage is misleading. The complicated nature of the different options that are combined in a convertible bond combined with the fact that it involves corporate bonds, which are by definition risky, makes it impossible to create a real risk-free strategy. However, given that this term is widely used in the hedge fund industry it will also be used throughout this paper.

chase performance or follow performance with some delay. The inflow of funds into convertible arbitrage hedge funds started increasing exponentially after the market downturn in 2000, when the returns on convertible arbitrage hedge fund indices peaked at more than 20%, but decreased sharply afterwards, turning negative in 2005. In case of the mutual fund industry, Berk and Green (2004) show that an increase in fund inflows is followed by a decrease in performance, caused by decreasing returns to scale. Since hedge funds are typically employing limits to the size of the assets they manage, we expect other factors to explain the decrease in returns (increased competition in the industry and market stability among others).

As the aim of the convertible arbitrage strategy is exploiting underpricing of convertible bonds, we first examine the pricing of convertible bond issues. Kang and Lee (1996) and Ammann, Kind, and Wilde (2003) provide evidence on convertible bond underpricing. The underpricing provides potential arbitrage opportunities, and may attract the attention of hedge funds, amongst others. Following Lewis, Rogalski, and Seward (1999) and Burlacu (2000) we employ the delta measure⁴ to make a distinction between debt and equity-like convertible bond issues. We use a sample of convertible bonds on the Canadian market issued in the period between 1998 and 2007. Based on the valuation model of Tsiveriotis and Fernandes (1998) we find the equity-like issues ($\Delta \geq 0.5$) about 27% underpriced, and the debt-like convertibles ($\Delta < 0.5$) about 7% underpriced at the issue. The underpricing does decline somewhat immediately following the issue, but nevertheless persists over a longer period of time. We show that the results are robust to finer delta buckets.

Second, by using information on aggregated bi-weekly short positions on the Toronto Stock Exchange (TSX) we examine changes in short positions (interest) around the issuance dates of convertible bonds. We observe a significant increase in the short positions (short interest) of the underlying stocks after the announcement of a convertible issue. In the 40 trading days following the announcement of the issue, the increases in relative short positions for equity-like issuers are about 15 percentage points higher than for debt-like issuers. These increased aggregated short positions remain stable after the issue of the convertible for a longer period of time. The Toronto Stock Exchange provides information on consolidated short positions twice a month, as reported by brokers. For comparison, until recently the data on short positions for the US market was only available on the monthly basis. Therefore, the bi-weekly Canadian data on consolidated short positions offers a better setting for the investigation of convertible arbitrage activities.

Finally, we show that increases in the short positions negatively affect the excess stock returns of the issuers, particularly during the period between the announcement and the issue of the convertible. Negative wealth effects associated with the increase in short positions have a direct negative effect for shareholders of the issuing companies. In addition to this direct effect, shareholders indirectly suffer from the lower market value of collateral (firm's assets), since this will negatively affect the credit spread of the existing and potential new debt claims (as shown by Longstaff and Schwartz, 1995). Through this same channel, debt claimants are directly negatively affected as well.

The realization of extreme gains and losses seem to have led to a change of the type and behavior of the issuers. We demonstrate that the issues became more debt-like over time making the convertible bond issues less underpriced at the issuance date and thus less

⁴ The delta is derived from the option pricing model of Black-Scholes (1973) and adjusted for continuous dividend payments in the way suggested by Merton (1973). Values closer to 1 indicate a high sensitivity of the convertible for the underlying stock value, implying a high probability of conversion.

attractive for convertible arbitrageurs. Furthermore, we argue that the negative effect on the underlying stock prices contributed to the shift of the universe of potential issuers or issue characteristics towards more debt-like convertible bonds in recent years. In our opinion this is an important explanation why convertible arbitrage returns declined in the recent years.

The remainder of the paper is structured as follows. In Section 2 we present the notion of convertible arbitrage and the role of hedge funds. In Section 3 we describe our sample data. This is followed by the main analysis regarding the pricing of convertibles and the short interest positions in Section 4. In Section 5 we analyze the relation between the short interest positions and their impact on the excess returns of issuer's stocks and discuss the reasons for the decline of convertible arbitrage returns in the recent past. Section 6 concludes.

2. Convertible Arbitrage and Hedge Funds

2.1 Convertible Arbitrage

The beginnings of convertible arbitrage, albeit not as refined and computationally sound as today, go back as far as the second half of the nineteenth century, when the first convertible securities were being issued (Calamos, 2005). The “arbitrage” setup was based on the same principle as today - taking a long position in bonds and a short position in the underlying stock. The specific number of shares of common stock to be sold short is a function of the conversion ratio (number of stocks into which the convertible bond converts), the sensitivity of the value of the conversion option to changes in the price of underlying equity (the delta measure), and the sensitivity of the delta measure to the changes in the price of underlying equity (the gamma measure).

The delta measure is defined as the change in the value of the conversion right due to the change in the value of the underlying equity. It can be derived from the option pricing model of Black and Scholes (1973), adjusted for continuous dividend payments in the way suggested by Merton (1973):

$$\Delta = \frac{\partial C}{\partial S} \quad (1)$$

$$\Delta = e^{-\delta T} \cdot N \left[\frac{\ln \frac{S}{K} + (r - \delta + \frac{\sigma^2}{2}) \cdot T}{\sigma \cdot \sqrt{T}} \right], \quad (2)$$

where C is the value of the conversion option, S is the current price of the underlying stock, K is the conversion price, δ is the continuously compounded dividend yield, r is the continuously compounded yield on a selected “risk-free” bond, σ is the annualized stock return volatility, T is the maturity of the bond and $N(\cdot)$ is the cumulative standard normal probability distribution. The delta measure always takes a value between 0 and 1. Values closer to 1 indicate a high sensitivity of the convertible bond value to the underlying equity (stock) value, implying a high probability of conversion.

The convertible arbitrage strategy provides the following cash flows: (1) cash inflows from coupon payments of the convertible, (2) cash inflows from the short interest credit on

the short stock account⁵, and (3) cash outflows due to dividend payments on shorted stock. This is also the reason why non-dividend paying stock is more desirable for the strategy. Finally, if at the time of the arbitrage setup the convertible bonds are underpriced, there is a potential for arbitrage profits.

The hedge ratio and the convertible arbitrage setup are time varying, since they depend on the stock price. When the stock price approaches the conversion price, the delta of a convertible bond increases – the bond becomes more equity-like (i.e. the price of the bond becomes more sensitive to the changes in the value of the underlying equity). This means that more stocks need to be shorted in order to maintain the neutral hedge ratio, which is defined as a product of the conversion ratio and delta. The opposite holds if the stock price goes down.

It should be noticed that the delta is not a perfect measure for the sensitivity of the conversion option to changes in the stock price. This is caused by the fact that the conversion option is in fact an option with a stochastic exercise price (since the underlying bond is used to pay for the exercise price). Besides that, convertible bonds are almost always callable, and sometimes also puttable. In addition, the exercise of a conversion option leads to the creation of new shares. All these features are not captured in the delta measure. However, given the fact that there is no better variable available than the delta measure, we continue to use this measure as an indicator of the sensitivity of the conversion option to changes in the price of the underlying stock.⁶

Calamos (2005) argues that convertible arbitrageurs in general look for convertible bonds that are more equity-like. The underlying shares have a higher volatility, which translates into a higher value of the equity option, a lower conversion premium and a higher gamma. Besides that, they have a preference for underlying stocks that pay low or no dividends, that are undervalued, liquid and that can be easily be sold short. Additionally, zero coupon convertible bonds or so-called LYONs (Liquid Yield Option Notes⁷) are said to be less desirable for convertible arbitrage, as they do not pay coupons and therefore lack cash inflows in the form of coupon components.

2.2 Convertible Arbitrage Hedge Funds

Convertible arbitrage has been one of the most successful hedge fund strategies of the end of the nineties and the beginning of 2000's. Using a survivorship free hedge fund dataset of Morningstar, we find that the number of convertible arbitrage hedge funds grew from about 28 in 1998 to about 121 in 2007. In the same period the assets under management grew from about 0.8 billion in January 1998 to about 12.9 billion in June 2004 and dropped to 9.4 billion in December 2007. The average annual return over the period 1998 - 2007 was 9.05% with an annual standard deviation of 4.35%. For comparison, during the same period the average total annual return of the S&P 500 was 6.84% with a corresponding standard deviation of 14.72%.

⁵The borrower of the stock (the party that short sells the stock) needs to keep a certain margin requirement with the broker where the shares were borrowed. This serves as a guarantee for the future return of the stock. This margin is typically about 50% of the value of the shorted stock. These funds are then credited with the short interest credit.

⁶For the same reason other studies use the delta measure as well; see e.g. Lewis, Rogalski and Seward (1999) or Dutordoir and Van de Gucht (2007).

⁷LYONs are zero coupon callable and puttable convertible bonds.

This indicates that the risk-reward trade-off for the convertible arbitrage strategy was better than that of a pure equity strategy.

According to Lhabitant (2002), convertible arbitrage trades represent more than half of the secondary market trading in convertible securities. This indicates that hedge funds are a very important liquidity provider in the convertible market. Hedge funds differ from mutual funds and other investment vehicles by their lack of regulation, with limited transparency and disclosure, and by their internal structure (see Fung and Hsieh, 1997). Most hedge funds try to achieve an absolute return target, irrespective of global market movements, while hedge fund managers typically have incentive-based contracts. Accordingly, hedge funds have a broad flexibility in the type of securities they hold and the type of positions they take. On the other hand, investors in hedge funds are often confronted with lockup periods and redemption notice periods. Such restrictions on withdrawals imply smaller cash fluctuations, and give fund managers more freedom in setting up long-term or illiquid positions.

The non-standard features make hedge funds an interesting investment vehicle for investors with potential diversification benefits. From an investor point of view, it appears that a convertible arbitrage strategy offers a significant diversification benefit due to a low correlation between a convertible arbitrage strategy and a pure equity index like the S&P500. During the period 1998 - 2007 this correlation was about 0.52. Using a sample of convertible bonds issued by US and Japanese firms in the period 1993-2003, Agarwal, Fung, Loon, and Naik (2008) show that most of the return variation in convertible arbitrage hedge fund indices can be explained buy-and-hedge strategy, where investors assume long position in convertible bonds and short position in the shares of convertible bond issuer.

3. Data

We investigate the convertible debt issues in the Canadian market between 1998 and 2007. Data regarding the issues and their characteristics is obtained from the SDC New Issues database and from prospectuses of the issuers (available on the SEDAR web site.⁸) Data on the stock prices, market indices, government bond yields, interest rates, dividends, and number of shares outstanding is obtained from Datastream. Data on convertible bond prices, their trading volumes, and number of trades is obtained from Stockwatch. Data on short interest (short sales) is obtained from the Toronto Stock Exchange Group (TSX Group). TSX provides the information on consolidated short positions for stocks traded on TSX and TSX Venture exchanges twice a month (every 15th and the last day of the month), as reported by brokers. Until recently, data on short positions for the US market for example was only available on monthly basis. Therefore, the bi-weekly Canadian data on consolidated short positions provides us with a better setting for examining patterns in the number of stocks sold short of the underlying equity of a convertible issue immediately after announcing and issuing the convertible. Moreover, short sales on the Canadian market are said to be easier (less limitation⁹) and less costly to execute than for example in the US market. This is especially the case for stocks of companies with options or convertible bonds outstanding. This makes

⁸SEDAR stands for "System for Electronic Document Analysis and Retrieval" and is a service of CSA (Canadian Securities Administration) providing public securities filings. (<http://www.sedar.com/>)

⁹According to Universal Market Integrity Rules set by Market Regulation Services of Canada (<http://www.rs.ca>), which replaced rules of individual exchanges in Canada, short sales are allowed on zero tick and in certain cases also on down tick.

the Canadian market a suitable setting for the investigation of short sales and convertible arbitrage returns.

3.1 Sample Selection and Description

In total, there were 174 new public convertible bond issues denominated in Canadian Dollars and registered in the SDC dataset during the period 1998-2007. We exclude all the exchangeable bonds¹⁰ and zero coupon bonds. In case of exchangeable bonds the options are not written on the issuer's equity, but rather on another asset (either other companies' stocks or a specific commodity). In this case there is an additional settlement risk involved, which the valuation model does not take into account. In the case of true convertibles, the issuer can always deliver its own equity (issue new shares), so that part of the convertible value is considered to be riskless.¹¹ Zero coupon bonds are excluded, because coupon payments are an important part of the cash flows for convertible arbitrageurs. Since zero coupon bonds do not provide these cash flows, they tend to be avoided by convertible arbitrageurs. We impose the requirement that announcement and issuance dates (completion of the offer) are verifiable either in company's announcements and prospectuses on the SEDAR website or in Lexis Nexis. Finally, all our bonds in the sample should have stock price and bond price data available on Datastream or Stockwatch, as well as all the details of the issue provided in the prospectus. This leaves us with a final sample of 101 convertible bond issues.

In Table 1 we present the descriptive statistics broken down by the year of the convertible bond issue.

<Insert Table 1 here>

We observe that changes in volatility and delta closely correspond over time. In particular, the average values of delta have decreased over time, from 0.613 in 1998 to 0.242 in 2007. This implies that, according to the delta measure, the average issue was much more equity-like at the beginning of our sample period than at the end. At the same time the average volatility of the issuer's stock price also decreased from 0.485 in 1998 to 0.368 in 2007. The dividend yield increased from 3.2% in 1998 to 11.6% in 2007. The average maturity of the issue declined from 8.5 years in 1998 to around 5.5 years in 2007.

The construction of the delta measure itself implies that lower volatility leads to lower delta values. The overall volatility in the market has declined after 2001, but we believe that the universe of issuers has also changed during the same period, thus additionally affecting the lower value of the delta. Given the important effect of volatility on the value of the delta, we investigate the changes in the volatility of the issuers by taking into account the changes in the market volatility at the same time. We look at the ratio between the volatility of the issuer at the time of the issue announcement and the market volatility at the same time ($\frac{\sigma_i}{\sigma_M}$). The higher the ratio, the riskier the issuer compared to the risk of the investment in the market index. In Table 1 we observe that the ratio was 3.262 in 1998, dropping over the years to

¹⁰Exchangeable bonds are bonds that are convertible into some other asset than the (equity) stock of the issuing company.

¹¹The valuation approach is explained in more detail in Section 4.1.

1.567 and 1.796 in 2003 and 2004 respectively and increasing to 4.361 and 2.978 in 2006 and 2007. This suggests that on average the issuers became less risky towards 2004 when the trend reversed and we can observe an increase in riskiness of the issuers in 2006 and 2007.

Next, we look at the conversion premium, which is defined as the difference between the conversion price and the stock price at the issue relative to the stock price $\left(\frac{K - S}{S}\right)$. The conversion premium (cp_i) is inversely related to the conversion ratio. Higher conversion ratios (lower conversion premiums) indicate more equity-like convertibles (Kim, 1990) and vice-versa, since a convertible bond with a lower conversion premium is more likely to become in-the-money (all else equal) and be converted into equity. Besides the maturity of the bond, the conversion ratio (or conversion price), on which the conversion premium depends, is the only parameter in Equation 2 which companies can choose. As shown in Table 1, the average conversion premium in our sample of convertible bonds has declined from 0.297 in 1998 to 0.164 and 0.186 in 2006 and 2007 respectively. All else equal, this would imply that convertible bonds have become more equity-like, which is in contrast to the conclusion based on the delta measure. However, we believe that the change in conversion premium indicates that issuers tried to offset the effect of lower volatility by lowering the conversion premium and making the issues more attractive for the investors. Otherwise, the issues would probably be more difficult to sell.

Finally, we investigate the industry composition of the issuers and observe significant changes in time.¹² Financial companies (SIC division H) accounted for between 40 to 55% of issuers in 1998 and 1999, with almost no issuers from SIC division A (agriculture, mining and construction) during the same period. In 2001 and 2002 we observe a decrease in the number of issues by financial companies (to one-third of the issues in a given year) and an increase in the number of issuers from SIC division D (manufacturing), which accounted for about one third of the issuers. In contrast, we observe 45 to 50% of the issues in 2003 and 2004 being made by companies in agriculture, mining and construction (SIC division A). This trend continued in 2005, while in 2006 and 2007 the majority of issues was placed by financial companies (income trusts).

In addition, we look at some of the other characteristics of the issue. Although the maturity of the bonds has decreased over time, the time-to-first call has not changed significantly, as it is mostly between 3 and 4 years over the whole sample period. Call-price premiums have increased, going from no premium at the end of the nineties to an average call-price premium of around 2 percent in 2004 and 2005. More interesting is the comparison of average coupons across the sample period. Although it seems that the average coupon has little variation over time (going from 7.1% in 1998 to 10.7% in 1999 and back to 6.6% by 2007), when the changes in the risk-free rate are taken into account, the relative coupon values (r_c) show more variation. Bonds in later years of the sample, which are also associated with a lower delta (more debt-like convertibles), seem to have higher credit spreads (r_c) than the more equity-like convertibles (higher delta). Although this might seem surprising at first (given the lower relative volatilities), it can be reconciled from the perspective of the valuation of convertibles. The equity part of the convertible bond is not risky in terms of its delivery, while the straight debt part bears the default risk. The larger the debt component, the higher the probability of default, and hence the riskier the bond. Finally, we look at the relative issues size (IS_i), defined as the ratio between the principal size of the issue and the market value of the issuing firm, which is measured as the average value over the 10-day

¹²We do not report complete results here. These results are available from the authors on request.

period prior to the announcement of the issue. We find that the changes in the relative issue size correspond to the changes in the delta, where the average issue size seems to decrease as the average delta increases (from 0.426 in 1999 to 0.168 in 2004 when the average delta value of the issue was at its lowest in the sample period).

We interpret the changes of the delta, volatility, conversion premiums and industrial composition as evidence for the fact that the universe of issuers and their characteristics has changed over time. This had an important impact on the delta measure (the design of the convertible bond issue), in addition to the effect of the overall lower volatility in the market.

4. The Cornerstones of the Convertible Arbitrage Strategy

The classic convertible arbitrage strategy involves taking a long position in a convertible bond and a short position in the underlying stock. In the previous section we demonstrated that the design of convertible bond issues has changed over time, i.e. the issues became more debt-like. In this section we want to examine whether there is a relation between the size of the delta of a convertible issue and the pricing of the bond and how it is related to changes in short interest in the underlying stock.

4.1 Pricing of Convertible Bonds

In general, a convertible bond can be considered as a bundle of a straight bond and a warrant written on the underlying equity. There are two theoretical approaches to valuing convertible debt. The so-called *structural models* use the value of the firm as the underlying state variable¹³, while in the so-called *reduced-form* models the value of the firm's equity or rather the default probability is modeled as underlying state variable.¹⁴ The reduced-form models have been adopted in most of the recent literature on the pricing of convertible debt.

Grimwood and Hodges (2002) argue that the most widely adopted model among practitioners for valuing convertible debt is the one first considered by Goldman Sachs in 1994 and later formalized by Tsiveriotis and Fernandes (1998), from now on TF. TF use a binomial tree approach to model the stock price process and decompose the total value of a convertible bond (CB) in an equity part and a straight debt part (so-called Cash Only part of a Convertible Bond - COCB). The holder of the hypothetical COCB receives all the cash flows, but no equity flows. The value of the COCB is determined by the convertible bond price, the underlying stock price, and the time to maturity, since these so-called early exercise parameters define the boundary conditions. In other words, since early call, put or conversion is possible, the stock prices that trigger these events represent the so-called free boundaries that affect the COCB and CB values. Since the COCB is risky, the partial differential equation (Black-Scholes) must include the issuer's risk or the credit spread to account for the relevant risk. The difference between the value of the convertible bond and COCB is the payment in equity. Since the firm can always deliver its own equity, this part can be discounted using the risk-free rate.¹⁵ In this paper we use the methodology of TF to estimate

¹³See for example Ingersoll (1977), Brennan and Schwartz (1977 and 1980), Nyborg (1996).

¹⁴See for example Jarrow and Turnbull (1995), Tsiveriotis and Fernandes (1998).

¹⁵For more details see Tsiveriotis and Fernandes (1998).

the model prices of convertible debt issues in our sample, since this approach can take into account any call, put, and conversion features of convertible debt.

In order to calculate the theoretical (model) price of a convertible bond we use the following inputs. For the risk-free rate we use the yield on government bonds (Canadian) of comparable maturity as the convertible bond. A static spread is used to correct for the credit risk of the issue. Where the data on credit risk (credit rating of the issue) are not available, we assumed that the company is of the BBB risk.¹⁶ In Datastream only Scotia Capital provides Canadian corporate bond benchmarks for different maturities and different credit ratings. They cover BBB, A, and AA rankings of short, medium, and long term. Based on the maturities we have extrapolated the following maturities: 1 year (equivalent to short term), 3 years (between short and medium term), 5 years (medium term), 7 years (between medium and long term), 10 years and more (long term). Based on the rankings, we have also extrapolated the rankings lower than BBB (BB and B) by adding a spread to BBB. This spread is relative to the spread between BBB and A, but is increasing in a lower credit quality and maturity. The price of the underlying stock at the valuation date is taken from Datastream, where we take the average stock price between days -12 and -2 relative to the announcement date of the issue. The number of steps in the tree is equal to the number of months to maturity at the issue of the bond. The coupon rate, the number of coupons per year, conversion ratio, and call schedule are all obtained from the respective prospectuses of the bonds. With respect to the dividend information, we obtained dividend yield data from Datastream.

We define mispricing as:

$$e_{i,t} = \frac{(M_{i,t} - B_{i,t})}{B_{i,t}}, \quad (3)$$

where for every issue i $e_{i,t}$ denotes the mispricing at time t , $M_{i,t}$ represents the model price at time t , computed using the approach to convertible bond valuation as previously described, and $B_{i,t}$ denotes the closing market price of a convertible bond at time t .

Based on the model and observed prices we investigate the mispricing of the convertible bond issues during the first year of trading. Volatility estimates¹⁷ are based on a rolling window of the past 250 trading days and the delta is estimated for every individual trading day. Both the risk-free rate and the credit spread are also considered for each trading day separately. Stock prices are matched to every individual trading day. We use constant dividend yields, computed as the average dividend yield of the past 250 trading days.

In Figure 1 we present the degree of mispricing ($e_{i,0}$) at the issuance date. It appears that the majority of the issues with a delta close to zero are underpriced, while a small number of issues with a delta of almost zero even show an overpricing of up to 6.7% (see the minimum value of underpricing for debt-like convertibles in Table 2).

¹⁶We have also computed model prices by taking the lowest possible credit quality for the issues with no credit risk information available. The mispricing was on average somewhat lower, but it did not significantly affect the results. These calculations are available on request from the authors.

¹⁷As a robustness check we also calculate model prices using implied volatilities for those convertibles that have underlying shares on which exchange-traded options are available. Unfortunately, implied volatilities are only available for five of the issues. For those issues we find an even larger underpricing if we use implied volatilities rather than historical volatilities. However, the difference between equity-like and debt-like convertibles is unaffected.

<Insert Figure 1 here>

For the convertible arbitrage strategy this would imply that a short position in the convertible has to be taken, combined with a long position in the underlying stock. In practice it will be very hard, if not impossible, to short a convertible, implying that hedge funds cannot benefit from this potential arbitrage opportunity. The issues characterized by higher delta values are all underpriced, with a maximum underpricing of about 50%. This implies that the underpricing of convertible debt issues is an increasing function of the delta of the issue. Based on this figure, we classified convertibles into more equity-like ($\Delta \geq 0.50$) and more debt-like issues ($\Delta < 0.50$). Additionally, we created a subsample of equity-like issues with $\Delta \geq 0.67$ and a subsample of debt-like issues with $\Delta < 0.33$. In Table 2 we present the degree of mispricing ($e_{i,t}$) at different points in time after the issue of the bond for the subsamples of the equity-like and the debt-like convertibles.

<Insert Table 2 here>

First, we observe that on average the equity-like convertibles ($\Delta \geq 0.5$) are underpriced by 26.8% at the issuance date, while the debt-like convertibles ($\Delta < 0.5$) are on average only underpriced by 7.0%. This difference of 19.8 percentage points is statistically significant. For the subsample of issues with a $\Delta \geq 0.67$ the underpricing is even more than 41%, while for the subsample of issues with a $\Delta < 0.33$ the underpricing is around 6%. The underpricing at the issuance date in the overall sample is on average 10%.¹⁸ These results are in line with the findings of Chan and Chen (2007), who find an 8% overall underpricing of convertibles at the issuance date in the US market.

Secondly, the underpricing on average declines in the first 5 trading days following the issue, with a decline of 3.2 percentage points in the case of the equity-like convertibles ($\Delta \geq 0.5$) and 1.2 percentage points in the case of the debt-like convertibles ($\Delta < 0.5$). It further decreases afterwards for the debt-like convertibles and remains at about 5%, while in the case of equity-like convertibles it shows a slight downward trend, but remains at around 19% by the end of the eleventh trading month following the issue. The difference in the underpricing between the equity-like and debt-like issues is still significant at 11.8 percentage points six months (120 trading days) after the issue. A similar result can be observed if we define equity- and debt-like convertibles based on the alternative cut-off values of delta – $\Delta \geq 0.67$ for the equity-like and $\Delta < 0.33$ for the debt-like convertibles.

Our findings are in line with the pricing of new equity and debt issues. It is a well founded fact in the financial literature that equity IPOs are underpriced. It is frequently argued that underpricing is a result of informational asymmetries, or liquidity risk, amongst some other factors. Datta, Iskandar-Datta, and Patel (1997) analyze pricing of corporate straight debt IPOs and find that speculative grade debt was similarly underpriced as equity IPOs. The reason is that the equity component (default risk) is much greater in the case of investment grade debt. Cai, Helwege, and Warga (2007) reach similar conclusions and additionally show that the underpricing is lower in the case of seasoned issues of debt. Corwin (2003) finds seasoned equity offerings to be underpriced, but the economic effect is much lower than in the case of equity IPOs.

¹⁸Note, that we do not report the total sample averages in detail, but these results are available on request.

4.2 Short Positions

One of the basic principles of convertible arbitrage is to short sell the underlying assets of the convertible bond, while purchasing the convertible bonds at the same time. For the purpose of investigating the relation between short sales, characteristics of the issue and underpricing we define a relative measure of short sales as the ratio between the short interest in a given period¹⁹ and the potential number of shares that are to be issued if the convertible bond issue is converted into shares²⁰:

$$Z_{i,t} = \frac{ss_{i,t}}{n_i^b \cdot cr_i} \quad (4)$$

$Z_{i,t}$ represents the relative short sales (interest) measure for company i at time t ($t=0$ is the announcement date), $ss_{i,t}$ represents the number of shorted shares (so-called short interest) of issuer i at time t , n_i^b denotes the number of issued bonds of issuer i and cr_i denotes the conversion ratio of the issue i . This measure of short interest standardizes outstanding short positions in every period with the number of new shares to be issued upon conversion of the convertible bond issue into the issuer's equity. After the announcement we expect Z_t (cross-sectional average at time t) to be significantly higher for the more equity-like convertibles ($\Delta \geq 0.5$ and $\Delta \geq 0.67$) than for the more debt-like convertibles ($\Delta < 0.5$ and $\Delta < 0.33$). The reason for this is that, given the convertible arbitrage strategy, more shares need to be sold short given the higher delta.

In Table 3 we present the descriptive statistics for both the level and the changes in the relative short sales between the consecutive periods in a cross-section of issuers at given points in time t following the announcement of the issue.

<Insert Table 3 here>

We observe that in the case of the more equity-like issues ($\Delta \geq 0.5$) the average relative short interest (Z_t) increases from around 5% in the last period before the announcement of the issue ($t=-1$) to 20.7% of the new potentially issued shares in four weeks ($t=2$) following the announcement date. In the case of the more debt-like convertibles ($\Delta < 0.5$) the mean relative short interest (Z_t) increases slightly from 9.2% prior to the announcement (at $t=-1$) to 12.0% in the period of the announcement ($t=0$). It declines to 9.6% in eight weeks following the announcement ($t=4$). The difference of 15.5 percentage points between the two sub-samples at $t=4$ (approximately eight weeks after the announcement of the issue) is statistically significant at the 1% level.²¹ The difference continues to increase following the issuance to 20 percentage points (at $t=7$) and then declines to 14.0 percentage points after 8 months ($t=16$) following the issue announcement. Even after 12 months ($t=24$) following the issue

¹⁹Note that the data on short interest (short positions) is available biweekly - in the middle and the beginning of every month.

²⁰We have also investigated the second relative measure of short sales defined as the ratio between the short interest in a given time period and the corresponding total number of shares outstanding. The results, which are available from the authors on request, are very similar and are only downscaled.

²¹ This result is in line with that of De Jong, Dutordoir, and Verwijmeren (2008) who also find a stronger increase in short sales on the issuance date for convertibles with higher deltas.

announcement, the mean relative short interest for the equity-like convertible issuers is 12.8 percentage points higher than the average short interest of the debt-like issuers. A similar result can be observed when comparing equity- versus debt-like issues based on delta buckets of $\Delta \geq 0.67$ and $\Delta < 0.33$ respectively.

In Panel B we present the results for the changes in relative short interest (dZ_t) relative to the last period before the announcement of the issue (Z_{-1}) and defined as $dZ_t = Z_t - Z_{t-1}$. From these results we conclude that the highest increase in the short interest for the equity-like convertibles ($\Delta \geq 0.5$) is at the announcement of the issue ($t=0$) and the immediate subsequent period $t=1$ (average increase of 4.6 and 7.7 percentage points respectively). This is followed by a more moderate increase of 3.4 percentage points in a period between two weeks and one month ($t=2$) after the announcement. Afterwards, the relative short interest keeps increasing, but at the lower pace of between 1 to 3 percentage points per two weeks. Contrary to that, companies that issue debt-like convertibles ($\Delta < 0.5$) experience an average 2.8 percentage points increase in short positions just after the announcement of the issue, and a 2.3 percentage points decline (almost a complete off-set) after the issue of the bond ($t=4$). A very similar evolution of short interest can be observed when comparing equity- versus debt-like issues based on delta buckets of $\Delta \geq 0.67$ and $\Delta < 0.33$ respectively.

5. Short Sales and Wealth Effects

As shown in Section 4, the short interest significantly increases after the announcements of convertible bond issues. Diamond and Verrechia (1987) argue that short interest negatively affects stock returns. Given the cost of shorting the stock, only informed traders will engage in the activity, thus conveying the new (bad) information to the market. A negative relation between short interest and abnormal returns is for example found by Aitken, Frino, McCorry, and Swan (1998) in the Australian market and Ackert and Athanassakos (2005) in the Canadian market.

We first investigate excess returns related to the period between the announcement and the issuance date of convertible debt issues.²² Previous literature²³ on the wealth effects typically focuses on a short event window around the announcement and / or issuance dates. In contrast to that, we investigate the wealth effects between the announcement and issuance dates of convertible debt issues, as we are interested in the evolution of abnormal returns during this period, and the corresponding pattern in the short positions of the underlying stocks. On average, the time between the announcement and issuance date is around 15 trading days, with a median value of 14 days and with 80% of the issues being in the range between 10 and 18 trading days.

<Insert Table 4 here>

²² We use the event study method to estimate the wealth effects associated with the issuance of convertible debt offerings. To estimate the parameters of the market model, we use the Standard & Poor's TSX (Toronto Stock Exchange) value-weighted total return index to compute the market return. It accounts for more than 200 stocks listed on the TSX or about 70% of the total market capitalization. The estimation period ranged from day -120 to day -20 relative to the announcement date of the offering.

²³ For a summary of previous empirical research on announcement effects see Loncarski, ter Horst, and Veld (2006).

From Table 4 we observe that some of the excess returns are significantly negative in the period between the announcement and the issuance date of convertibles. Equity-like convertible bond issuers ($\Delta \geq 0.50$ and $\Delta \geq 0.67$) experience a significant 6%-7% negative CAAR in the 15 day period leading to the issuance date versus around 1% negative CAAR for the debt-like convertible issuers ($\Delta < 0.50$ and $\Delta < 0.33$). These results are in line with the findings of Arshanapalli et al. (2004), who investigate announcement and issuance date wealth effects for a sample of 229 convertible issues in the U.S. market in the period between 1993 and 2001. They document a significant negative CAAR of 3.8% during the period of five days leading to the convertible issue. Note that they do not investigate the entire period between the announcement and the issuance date and that they do not differentiate between the more equity-like and the more debt-like convertibles.

The relation between short sales, structure of the convertible bond issues, and underpricing yields evidence that the more equity-like issues are more underpriced, and thus more interesting for arbitrageurs. Since in such cases more stock has to be sold short, we expect more downward pressure on the cumulative average abnormal returns (excess returns) between the announcement date and the issuance date.²⁴ We include an interaction term between a dummy variable for the type of the issue (equity- versus debt-like) and changes in short interest in order to capture differential effect of the structure of the issue on the changes in short interest on the excess returns. Although the delta measure itself, which we use to estimate the structure of the issue, is defined continuously on the interval between 0 and 1, the empirical distribution in our sample exhibits a certain degree of clustering at the very low and high values of delta. Additionally, in order to control for the dilution effect of the issue, we include the relative issue size in the regression specification. Therefore, we examine the relation between cumulative average abnormal returns prior to the issuance date and relative short interest by estimating the following regression model:

$$CAAR_{(t_1, ID), i} = \beta_0 + \beta_1 Z_{i,-1} + \beta_2 dZ_{i,0} + \beta_3 dZ_{i,1} + \beta_4 D_{\Delta} dZ_{i,0} + \beta_5 D_{\Delta} dZ_{i,1} + \beta_6 IS_i + \epsilon_i, \quad (5)$$

$CAAR_{(t_1, ID), i}$ represents cumulative average abnormal returns for the issuer i between time t_1 and the issuance date, $Z_{i,-1}$ represents relative short interest for the issuer i in the period before the announcement of the convertible issue, $dZ_{i,0}$ and $dZ_{i,1}$ represent the increase in short interest for the issuer i in periods 0 (around the announcement of the issue) and 1 (just before the issue of the convertible), D_{Δ} represents a dummy variable that takes value 1 for the more equity-like convertibles ($\Delta \geq 0.50$) and value 0 for the more debt-like convertibles ($\Delta < 0.50$) and IS_i represents the relative issue size, defined as the ratio between the principal size of the issue and the market value of the issuer's equity, which is computed as the 10-day average prior to the announcement of the issue ($\frac{\text{principal}_i}{\text{market value}_i}$). In Table 5 we present estimates for four different event windows between the announcement and the issue of the convertible bond.²⁵

²⁴ Loncarski, ter Horst, and Veld (2008), for example, find a negative downward pressure on cumulative abnormal returns that goes beyond the announcement date effects of the convertible bond issue and continues up to the issuance of the bond.

²⁵ Note that we have trimmed the sample to 80% of the observations based on the length of the period between the announcement and the issue date (top and bottom 10% were cut). This was done in order to homogenize the length of this period (varies between 0 to 40 trading days), as we are interested in the period between the announcement date of the issue and the issuance date.

<Insert Table 5 here>

The longest event window that we consider is between 15 days prior to the issue date of the bond and the issue date. This corresponds to the time period after the announcement, in which the announcement effect has been absorbed in the stock price, and the issue of the bond. We observe no relation between changes in relative short interest in the period of the announcement of the issue ($dZ_{i,0}$) and the cumulative average abnormal return ($CAAR_{(t_1,TD),i}$) for the debt-like convertibles in different time periods up to the issue of the bond. However, this relation is significantly negative in the case of equity-like convertibles (significantly negative coefficient for $D_{\Delta}dZ_{i,0}$). We concentrate on the period between 10 days prior to the issue and the issuance date. Here, we observe a marginally significant negative effect of the pre-announcement level of the short interest (coefficient of -0.061) and a significant negative effect of the change in the short interest in the period of the announcement for the equity-like convertibles (coefficient of -0.707). This implies that for the equity-like convertibles the average change in relative short interest in the period of the issue announcement ($dZ_{i,0}$) for the equity-like convertibles ($\Delta \geq 0.5$) of 4.6% (see Panel B of Table 3) corresponds to a decline in the CAAR of 3.25% in the period between 10 days prior to the issue and the issuance date. We find no significant effect of the relative issue size on the excess returns.

Convertible arbitrage activities generally have a positive effect for hedge funds and other investors that engage in them. However, these returns seem to come at the expense of “insiders” of the issuing firms. Short selling has a negative effect on abnormal returns - a direct negative effect for shareholders of the issuing companies. The results imply that convertible arbitrage activities might create additional (apart from the announcement date effect) adverse wealth effects for shareholders of the issuing companies. One might argue that such a result can simply be attributed to other activities of investors, such as the valuation shorting – i.e. equity-like convertibles are a substitute for equity issues. As such, similar to the case of seasoned equity offerings (issues), investors short the overvalued equity at the announcement of the issue. Loncarski (2007) analyzes changes in short interest for straight debt issuers, equity issuers, convertible bond issuers, and firms that repurchase shares in the Canadian market. They find the largest changes in short interest for convertible bond issuers, followed by equity issuers. Moreover, while changes in the short interest for equity issuers are of a short-term nature, implying more valuation shorting, changes in the short interest for convertible bond issuers are of a long-term nature, implying other activities (arbitrage) aside from valuation shorting. If there was no convertible arbitrage, there would be less increase in short interest (sales) and no or less additional negative wealth effect, in particular for the issuers of more equity-like convertibles. In addition, some other studies have also found that arbitrage activities create additional downward pressure on stock prices (see for example Mitchell, Pulvino and Stafford (2004) for merger arbitrage). In our opinion, all this evidence suggests that in case of convertible arbitrage investors short stocks immediately following the announcement of the issue in a large part as a result of arbitrage activities.

The additional downward pressure on excess returns caused by the increase in short interest has consequences for both shareholders and debtholders of the issuing company. It clearly negatively affects shareholders by additionally destroying shareholder value. Moreover, it also affects debtholders. Longstaff and Schwartz (1995) develop a model in which they show that value of firm’s assets (proxied by stock market index) negatively affects

credit spread. This implies that existing debtholders are affected by the negative wealth effect as the lower value of firm's collateral generally leads to an increase in the credit spread and thus a lower value of current (outstanding) debt claims. In such a context convertible arbitrage seems to have a negative effect for the shareholders and debtholders of the issuing companies, in particular when the issue is structured to be more equity-like (high value of delta). In our opinion this extends the explanation of deteriorating performance of convertible arbitrage in recent years, as the universe of issues (and issuers) shifted towards more debt-like issues, where lower increases in short positions create less negative pressure on the excess returns.

It should be emphasized that we are not implying that convertible arbitrageurs are somehow responsible for the negative CAARs. These effects are first and foremost caused by the underlying companies issuing underpriced convertibles. The convertible arbitrageurs only try to profit from this underpricing. In this context it is also interesting to mention the recent trend of companies issuing convertibles and simultaneously repurchasing stock (see De Jong et al. (2008) and Verwijmeren (2008)). This combination, which according to Verwijmeren (2008) is referred to as a "happy meal" by practitioners, is intended to facilitate convertible arbitrageurs who buy up the convertibles. By buying back their own stock, the companies counteract the price pressure effects of convertible arbitrage short selling activities.

6. Conclusion

This paper studies the effects of convertible bond arbitrage strategy on capital markets and its participants and provides an additional explanation of diminishing returns that the strategy provided in recent years.

Convertible arbitrage mostly takes place by hedge funds. While it has been one of the most successful hedge fund strategies in the past, consistently providing double digit returns, its performance has deteriorated in recent years. The popular press suggests that this "demise" can be attributed to stable equity markets, rising interest rates, withdrawals from funds, increased competition in the hedge fund industry, and lower volatilities in the main capital markets. We argue that an obvious shift in the universe of the issuers and the structure of the issues is important additional explanatory factor. We show that the convertible bonds are underpriced and that short positions in the underlying stocks strongly increase around the issuance date. Both effects are stronger for equity-like than for debt-like convertible bonds. Moreover, we demonstrate that increases in short positions, which are also attributable to arbitrage activities, negatively affect abnormal returns around the issue date of convertibles. This adverse effect is especially pronounced in the case of equity-like issues, as more stock is sold short. This has important negative implications for both shareholders and existing debtholders of the issuing companies. In order to cap these losses in shareholder value, issuers apparently either switched to issuing less underpriced debt-like convertible bonds, packaged the convertible bonds issues together with share repurchases and/or they moved out of the convertible bond market altogether. In our opinion, this shift provides an additional explanation for the lower returns on the convertible arbitrage strategy in recent years.

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Table 1: Descriptive Statistics

Descriptive statistics (number of observations, mean and standard deviation) for maturity (in years), delta, issuer and market volatilities, conversion premium, dividend yield, coupon rate, relative coupon rate, time-to-first-call and call-price premium. The statistics are provided based on the year of the issue. T represents the maturity of the bond at the issue. Both, issuer and market volatility (σ_i and σ_M) are the annualized standard deviations of daily stock returns and returns on S&P TSX composite index based on 250 trading days prior to the announcement of the convertible bond issue. The delta measure (Δ_i) is computed as defined in Equation 2. The conversion premium (cp_i) is defined as $\frac{K-S}{S}$, where K represents the conversion price and S represents the stock price at the announcement date of the issue. The dividend yield (δ_i) is the average dividend yield over the 250 trading days prior to the announcement of the issue. c_i represents the coupon rate of the bond. rc_i denotes the relative coupon rate, defined as the difference between the coupon rate of the bond and the “risk-free” rate, which is proxied by the yield of the government bond of the comparable maturity. The time-to-first-call (T_{call}) is the time between the issue of the bond and the first possible call date expressed in years. The call-price premium (p_{call}) is defined as the ratio between the call price and the par value of the bond, less one ($\frac{P_{call}}{P_{par}} - 1$). The relative issue size (IS_i) is defined as the ratio between the principal size of the issue and the market value of the issuer’s equity, which is computed as the 10-day average prior to the announcement of the issue ($\frac{\text{principal}_i}{\text{market value}_i}$).

year		T	Δ_i	σ_i	σ_M	$\frac{\sigma_i}{\sigma_M}$	cp_i	δ_i	c_i	rc_i	T_{call}	p_{call}	IS_i
1998	n	9	7	9	9	9	8	7	9	9	8	8	9
	mean	8.49	0.613	0.485	0.144	3.262	0.297	0.032	0.071	0.013	4.04	0.000	0.250
	std. dev.	2.24	0.287	0.417	0.017	2.626	0.159	0.052	0.016	0.015	1.66	0.000	0.110
1999	n	5	4	5	5	5	4	4	5	5	4	4	3
	mean	4.61	0.405	0.441	0.196	2.264	0.174	0.099	0.107	0.050	3.13	0.000	0.426
	std. dev.	1.67	0.271	0.179	0.004	0.956	0.163	0.076	0.012	0.017	0.08	0.000	0.025
2001	n	7	7	7	7	7	7	7	7	7	7	7	7
	mean	5.68	0.613	0.548	0.211	2.685	0.077	0.035	0.076	0.020	3.29	0.000	0.208
	std. dev.	0.93	0.330	0.367	0.025	1.826	0.100	0.052	0.010	0.009	0.40	0.000	0.119
2002	n	12	12	12	12	12	12	12	12	12	12	12	12
	mean	5.14	0.368	0.389	0.167	2.310	0.233	0.085	0.089	0.037	3.58	0.004	0.232
	std. dev.	0.17	0.286	0.225	0.017	1.284	0.3	0.072	0.006	0.009	1.08	0.014	0.098
2003	n	11	11	11	11	11	11	11	11	11	10	10	11
	mean	6.26	0.239	0.228	0.143	1.567	0.165	0.090	0.078	0.030	3.18	0.010	0.157
	std. dev.	1.78	0.267	0.142	0.021	0.854	0.174	0.055	0.014	0.016	0.63	0.021	0.067
2004	n	17	17	17	17	17	17	17	17	17	16	16	19
	mean	6.73	0.111	0.187	0.104	1.796	0.198	0.118	0.073	0.026	3.44	0.022	0.168
	std. dev.	1.94	0.093	0.038	0.008	0.383	0.394	0.041	0.009	0.014	0.48	0.026	0.086
2005	n	9	9	9	9	9	9	9	9	9	6	6	9
	mean	8.79	0.118	0.245	0.097	2.520	0.182	0.107	0.064	0.022	5.37	0.021	0.185
	std. dev.	8.17	0.261	0.101	0.008	0.918	0.349	0.046	0.010	0.011	0.97	0.025	0.134
2006	n	22	17	19	22	19	19	22	22	19	3	3	20
	mean	5.94	0.235	0.533	0.123	4.361	0.164	0.075	0.068	0.024	3.22	0.000	0.245
	std. dev.	1.29	0.337	0.810	0.008	6.601	0.299	0.060	0.009	0.007	0.09	0.000	0.165
2007	n	9	9	9	9	9	9	9	9	9	7	7	9
	mean	5.54	0.242	0.368	0.124	2.978	0.186	0.116	0.066	0.022	3.24	0.000	0.301
	std. dev.	0.88	0.284	0.106	0.004	0.862	0.060	0.054	0.007	0.007	0.38	0.000	0.288

Table 2: Convertible Bond Mispricing

Descriptive statistics (mean, minimum, and maximum value) for the mispricing for the different sub-samples of convertibles based on the value of delta and the difference in means between the equity-like and the debt-like convertibles based on different cut-off values for delta. Mispricing is defined as $e_{i,t} = \frac{(M_{i,t} - B_{i,t})}{B_{i,t}}$, $M_{i,t}$ represents the model price and $B_{i,t}$ represents the trading price at time t. ID+t denotes t days after the issuance date of the convertible bond. ***, ** and * denote significance at the level below 1, 5 and 10% respectively. Under the null hypothesis means and differences in means are equal to zero.

Period	N	mean	min	max	N	mean	min	max	N	mean	min	max	N	mean	min	max	Difference in means							
	$\Delta < 0.33$				$\Delta < 0.50$				$\Delta \geq 0.50$				$\Delta \geq 0.67$				$\Delta \geq 0.50 - \Delta < 0.50$		$\Delta \geq 0.67 - \Delta < 0.33$					
ID	61	0.063	***	-0.067	0.337	65	0.070	***	-0.067	0.337	16	0.268	***	0.021	0.500	9	0.419	***	0.119	0.500	0.198	***	0.357	***
ID+5	63	0.054	***	-0.078	0.278	68	0.058	***	-0.078	0.278	19	0.236	***	-0.008	0.471	10	0.386	***	0.086	0.471	0.178	***	0.332	***
ID+10	63	0.047	***	-0.115	0.269	68	0.051	***	-0.115	0.269	19	0.221	***	-0.020	0.463	10	0.373	***	0.073	0.463	0.170	***	0.326	***
ID+15	63	0.038	***	-0.121	0.258	68	0.043	***	-0.121	0.258	19	0.211	***	-0.038	0.473	10	0.345	***	0.054	0.473	0.168	***	0.306	***
ID+20	63	0.039	***	-0.105	0.304	68	0.046	***	-0.105	0.304	20	0.202	***	-0.017	0.472	11	0.401	***	0.028	0.472	0.156	***	0.362	***
ID+30	66	0.043	***	-0.090	0.286	71	0.050	***	-0.090	0.286	20	0.190	***	-0.026	0.514	11	0.320	***	0.030	0.514	0.140	***	0.277	***
ID+60	66	0.046	***	-0.067	0.310	71	0.052	***	-0.067	0.310	21	0.198	***	-0.019	0.499	12	0.331	***	0.012	0.499	0.146	***	0.285	***
ID+120	67	0.057	***	-0.091	0.514	72	0.067	***	-0.091	0.514	22	0.184	***	-0.069	0.376	12	0.305	***	0.054	0.376	0.118	***	0.248	***
ID+220	67	0.047	***	-0.150	0.341	72	0.062	***	-0.150	0.516	22	0.196	***	-0.117	0.848	12	0.336	**	-0.017	0.848	0.134	***	0.289	***

Table 3: Levels and Changes in Short Interest

Descriptive statistics (mean, minimum, and maximum) for the relative measure of short interest for the sub-samples of equity- and debt-like convertibles. The relative short interest is defined as $Z_{i,t} = \frac{ss_i}{n_i^b \cdot cr_i}$, where $Z_{i,t}$ represents the relative short sales (interest) measure for company i at time t (t=0 is the short interest reporting period at the announcement date of the issue), $ss_{i,t}$ represents number of shorted shares of issuer i at time t, n_i^b denotes the number of issued bonds of issuer i and cr_i denotes the conversion ratio of the issue i. Δ_i denotes the delta value of the issue. ***, ** and * denote significance at the level below 1, 5 and 10% respectively. Under the null means are equal to zero.

Panel A: Relative Short Interest for the different Sub-samples of Convertibles Based on the Value of the Delta

Period	N	mean	min	max	N	mean	min	max	N	mean	min	max	N	mean	min	max	Difference in means		
	$\Delta < 0.33$				$\Delta < 0.50$				$\Delta \geq 0.50$				$\Delta \geq 0.67$				$\Delta \geq 0.50 - \Delta < 0.50$	$\Delta \geq 0.67 - \Delta < 0.50$	
t=-2	61	0.075	0.000	0.770	71	0.082	0.000	0.770	23	0.043	0.000	0.326	12	0.016	0.000	0.082	-0.039	*	-0.059 ***
t=-1	61	0.086	0.000	0.498	71	0.092	0.000	0.598	23	0.050	0.000	0.360	12	0.014	0.000	0.089	-0.042	*	-0.072 ***
t=0	61	0.118	0.000	0.787	71	0.120	0.000	0.787	23	0.096	0.000	0.368	12	0.071	0.000	0.208	-0.024		-0.047 *
t=1	61	0.124	0.000	0.872	71	0.126	0.000	0.872	23	0.173	0.000	0.664	12	0.130	0.000	0.400	0.047		0.006
t=2	61	0.120	0.000	0.980	71	0.122	0.000	0.980	23	0.207	0.000	0.780	12	0.170	0.000	0.491	0.085	*	0.050
t=3	61	0.117	0.000	1.681	71	0.119	0.000	1.681	23	0.224	0.000	0.800	12	0.189	0.000	0.549	0.104	**	0.072
t=4	61	0.091	0.000	0.640	71	0.096	0.000	0.640	23	0.251	0.000	0.952	12	0.240	0.000	0.839	0.155	***	0.149 **
t=5	61	0.111	0.000	1.203	71	0.113	0.000	1.203	23	0.269	0.000	1.128	12	0.254	0.000	0.843	0.156	***	0.143 **
t=6	61	0.110	0.000	1.269	71	0.112	0.000	1.269	23	0.273	0.000	1.119	12	0.264	0.000	0.855	0.161	***	0.154 **
t=7	61	0.095	0.000	0.702	71	0.099	0.000	0.702	23	0.299	0.000	1.320	12	0.327	0.000	1.320	0.200	***	0.231 **
t=8	61	0.098	0.000	0.660	71	0.102	0.000	0.660	23	0.277	0.000	0.971	12	0.297	0.000	0.927	0.175	***	0.198 **
t=16	61	0.113	0.000	0.999	71	0.115	0.000	0.999	23	0.255	0.000	0.788	12	0.257	0.021	0.702	0.140	***	0.144 **
t=24	61	0.107	0.000	0.478	71	0.106	0.000	0.478	23	0.234	0.002	0.813	12	0.208	0.012	0.563	0.128	***	0.102 **

Panel B: Changes in the Relative Short Interest for the Different Sub-samples of Convertibles Based on the Value of the Delta

Period	N	$dZ_t = Z_t - Z_{t-1}$	N	$dZ_t = Z_t - Z_{t-1}$	N	$dZ_t = Z_t - Z_{t-1}$	N	$dZ_t = Z_t - Z_{t-1}$
	$\Delta < 0.33$		$\Delta < 0.50$		$\Delta \geq 0.50$		$\Delta \geq 0.67$	
t=-1	61	0.011	71	0.010	23	0.007	12	-0.002
t=0	61	0.032	71	0.028	23	0.046	12	0.057
t=1	61	0.006	71	0.006	23	0.077	12	0.059
t=2	61	-0.004	71	-0.004	23	0.034	12	0.040
t=3	61	-0.004	71	-0.003	23	0.017	12	0.019
t=4	61	-0.026	71	-0.023	23	0.027	12	0.051
t=5	61	0.020	71	0.017	23	0.018	12	0.014
t=6	61	-0.001	71	-0.001	23	0.004	12	0.010
t=7	61	-0.015	71	-0.013	23	0.026	12	0.062
t=8	61	0.003	71	0.002	23	-0.022	12	-0.030

Table 4: Excess Returns Around Issuance Date of Convertibles

Excess returns associated with the issuance dates of convertible bond issues for subsamples of equity- and debt-like convertibles based on the delta measure. The cumulative average abnormal return (CAAR) window is relative to the issuance date. The difference in means is computed as a one-sided test where, under the null hypothesis, CAAR for the more debt-like convertibles ($\Delta < 0.33$ and $\Delta < 0.50$) are less or equal to CAAR for the more equity-like convertibles ($\Delta \geq 0.67$ and $\Delta \geq 0.50$ respectively). ***, ** and * denote significance at the level below 1, 5 and 10% respectively. Under the null means are equal to zero.

CAAR window		$\Delta_i < 0.33$		$\Delta_i < 0.5$		$\Delta_i \geq 0.5$		$\Delta_i \geq 0.67$		$\Delta \geq 0.50 - \Delta < 0.50$	$\Delta \geq 0.67 - \Delta < 0.33$
		n	CAAR	n	CAAR	n	CAAR	n	CAAR	Diff. in CAAR	Diff. in CAAR
-15	Issuance date	58	-0.0148 **	67	-0.0121 **	23	-0.0710 ***	12	-0.0596 ***	-0.0589 **	-0.0448 **
-10	Issuance date	58	-0.0082 *	67	-0.0065	23	-0.0164	12	-0.0246	-0.0099	-0.0164
-5	Issuance date	58	-0.0050	67	-0.0038	23	-0.0165 *	12	-0.0021	-0.0127	0.0028
	Issuance date	58	-0.0021	67	-0.0016	23	-0.0077	12	-0.0093	-0.0061	-0.0072

Table 5: Excess Returns and Short Interest

Estimation results for the OLS regression model (see equation 5). The dependent variable is the cumulative average abnormal return (CAAR) between time t_1 and t_2 for different values of t_1 and t_2 where $t = 0$ represents the issuance date of the convertible bond issue. $Z_{i,-1}$ represents the relative short interest (as defined in equation 4) for the issuer i in the short interest reporting period before the announcement of the convertible issue, $dZ_{i,0}$ and $dZ_{i,1}$ represent the change in the short interest for the issuer i in periods 0 (around the announcement of the issue) and 1 (around the issue of the convertible) compared to the previous period, D_{Δ} represents a dummy variable that takes value 1 for the more equity-like convertibles ($\Delta > 0.50$) and value 0 for the more debt-like convertibles ($\Delta < 0.50$) and IS_i represents relative issue size, defined as the ratio between the principal size of the issue and the market value of the issuer's equity, which is computed as the 10-day average prior to the announcement of the issue ($\frac{\text{principal}_i}{\text{market value}_i}$).

***, ** and * denote significance at the level below 1, 5 and 10 percent respectively. Under the null hypothesis regression coefficients are equal to zero.

variable	coeff.	s.e.	t-value	p-value	adj. R^2
$t_1 = -15, t_2 = \text{Issuance Date}$					
$Z_{i,-1}$	-0.055	0.051	-1.09	0.280	
$dZ_{i,0}$	-0.125	0.121	-1.04	0.304	
$dZ_{i,1}$	0.022	0.077	0.28	0.778	
$D_{\Delta}dZ_{i,0}$	-0.694	0.276	-2.52	0.014	**
$D_{\Delta}dZ_{i,1}$	0.232	0.279	0.83	0.409	
IS_i	0.020	0.014	1.47	0.146	
constant	-0.020	0.013	-1.48	0.144	0.035
$t_1 = -10, t_2 = \text{Issuance Date}$					
$Z_{i,-1}$	-0.061	0.033	-1.82	0.074	*
$dZ_{i,0}$	-0.010	0.077	-0.13	0.901	
$dZ_{i,1}$	-0.006	0.052	-0.12	0.903	
$D_{\Delta}dZ_{i,0}$	-0.707	0.219	-3.23	0.002	***
$D_{\Delta}dZ_{i,1}$	0.310	0.216	1.43	0.157	
IS_i	0.006	0.009	0.65	0.517	
constant	-0.002	0.009	-0.28	0.782	0.042
$t_1 = -5, t_2 = \text{Issuance Date}$					
$Z_{i,-1}$	-0.017	0.030	-0.55	0.582	
$dZ_{i,0}$	-0.079	0.087	-0.92	0.363	
$dZ_{i,1}$	-0.015	0.035	-0.43	0.667	
$D_{\Delta}dZ_{i,0}$	-0.353	0.175	-2.02	0.048	**
$D_{\Delta}dZ_{i,1}$	0.012	0.011	1.16	0.251	
IS_i	0.012	0.014	0.84	0.403	
constant	-0.007	0.008	-0.92	0.362	0.044
Issuance Date					
$Z_{i,-1}$	0.003	0.011	0.24	0.813	
$dZ_{i,0}$	-0.021	0.051	-0.42	0.679	
$dZ_{i,1}$	-0.021	0.019	-1.09	0.281	
$D_{\Delta}dZ_{i,0}$	-0.073	0.203	-0.36	0.721	
$D_{\Delta}dZ_{i,1}$	0.091	0.167	0.54	0.589	
IS_i	0.002	0.004	0.42	0.673	
constant	-0.002	0.003	-0.53	0.600	-0.067

Figure 1: Relation between Mispricing and the Delta at the Issuance date of the Convertible Bond

Values of the delta and the degree of mispricing at the issuance date $e_{i,0}$. The delta measure ($\Delta_{i,0}$) is computed at the issuance date as defined in Equation 2. Mispricing is defined as $e_{i,0} = \frac{(M_{i,0} - B_{i,0})}{B_{i,0}}$, where $M_{i,0}$ and $B_{i,0}$ represent the model and the trading price at the issuance date respectively.

